Meaning-preserving text simplification by activation steering

Technical report

30 July 2026

Result. One dial, structural, layer 15, $\alpha = 0.75$.

vs. prompt-only, SARI	+5.47	[+2.45, +8.49]
vs. prompt-only, NLI	+0.36	[+0.23, +0.50]
vs. no intervention, SARI	+0.43	[−1.75, +2.61]
vs. no intervention, NLI	−0.08	[−0.21, +0.05]
sentence length		−18%
degeneration		0%

Steering beats prompting and matches no-intervention. Prompting is *worse* than no intervention at −5.04 SARI.

Abstract

We steer a 3B instruction-tuned language model toward simpler text by adding a fixed direction to its residual stream during generation. Two candidate axes are tested: sentence structure and word difficulty. The axes are separable directions, mean absolute cosine 0.223, and they respond at different depths: structure at layers 15 to 17 of 28, vocabulary at layers 19 to 26. Structural steering at layer 15 with strength 0.75 beats prompt-only simplification on ASSET by SARI +5.47 (95% CI +2.45 to +8.49) and by bidirectional-NLI faithfulness +0.36 (CI +0.23 to +0.50), both significant at n=32. Against no intervention it is a SARI tie (+0.43, CI −1.75 to +2.61) with faithfulness unchanged, while cutting sentence length 18% at zero degeneration. Prompt-only simplification is significantly worse than no intervention (−5.04 SARI): it reaches a low readability grade by deleting and inventing content. The lexical axis does not pay for itself, so the deliverable is one dial. We also report a floating-point defect on Apple MPS that silently corrupted every extracted vector and produced a false negative result, and two metric defects that inverted conclusions.

1 Problem

Readers who are dyslexic, cognitively impaired, or new to a subject need complex text rewritten simply. The rewrite must not change what the text says. Dropping a qualifier or inventing an example defeats the purpose.

Prompting an instruction-tuned model to simplify is a switch. It fires or it does not. Different readers need different degrees of simplification, and a teacher cannot ask one prompt for grade 4 and grade 7 simultaneously.

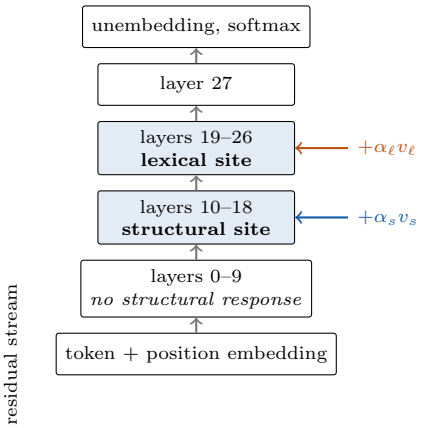


Figure 1: Injection sites in the 28-layer stack. Each decoder layer adds to the residual stream; steering adds one more term. The two depth bands are the measured response regions from Section 9, not a design choice.

Activation steering supplies a continuous parameter instead. A direction is extracted offline and added to the model's internal state at generation time, scaled by a strength α. That parameter is the object of interest.

2 Method

A decoder-only transformer maintains a residual stream: a vector per token position, updated additively by each layer. Steering modifies that stream in place. For layer l and steering vector v,

$$h_l \leftarrow h_l + \alpha v, \tag{1}$$

optionally followed by rescaling h_l to its original norm. Figure 1 shows where the intervention sits.

Injection is implemented with PyTorch forward hooks on the decoder-layer modules. A hook receives the layer output and may return a replacement. The recorder counterpart reads the same tensor and returns `None`, which leaves the output untouched.

Position selection matters. With a key-value cache, prefill is one call carrying the whole prompt and each decode step carries one token. Without a cache, every call carries the whole sequence. Injecting on

generated tokens only therefore cannot be decided by counting calls; the implementation compares the current sequence length against the remembered prompt length.

Setup. Apple M4 Max, 128 GB unified memory, bfloat16 on the MPS backend. Model `Llama-3.2-3B-Instruct`: 28 layers, hidden size 3072. Faithfulness scoring uses `roberta-large-mnli`.

Weights are fetched with `aria2c` at 16 connections per file and 4 files concurrently, because the Hugging Face client opens one connection per file. One access detail cost time: `meta-llama` gates Llama-3.2 behind a licence separate from Llama-3.1, so a token can read Llama-3.1 and receive HTTP 403 on every Llama-3.2 file while repository *metadata* stays public. Listing files succeeds and downloading fails. The ungated mirror `unsloth/Llama-3.2-3B-Instruct` carries byte-identical bfloat16 shards (4 965 799 096 and 1 459 729 952 bytes), verified against the gated repository's own metadata.

3 Extracting vectors

Three estimators produce the same output shape, so the sweep can compare them.

Instruction difference. Take a corpus of complex texts. For each, run two forward passes whose prompts differ only in one instruction, and average the difference of the pooled hidden state:

$$v_l = \frac{1}{N}\sum_{i=1}^{N}\big[h_l(x_i \oplus q) - h_l(x_i \oplus \bar{q})\big]. \qquad (2)$$

Here q is the axis instruction and $\bar{q}$ its contrast. Both prompts end in the same generation-prompt token, so pooling the last position isolates the instruction's effect on the state from which the model begins to answer.

The choice of $\bar{q}$ decides whether this works. Setting $\bar{q} = \emptyset$ fails: the difference is then dominated by the prompt being longer, not by what the instruction said. Measured on 256 texts, the structural and lexical vectors came out at mean absolute cosine 0.983 under that contrast. Replacing $\bar{q}$ with an axis-matched placebo of comparable length cancels the shared component. The structural pair is "Use short, simple sentences with one idea each" against "Keep the original sentence structure and sentence length exactly as they are"; the lexical pair varies wording and holds structure fixed.

Contrastive activation addition. Given aligned pairs,

$$v_l = \mathbb{E}[h_l(\text{simple})] - \mathbb{E}[h_l(\text{complex})]. \qquad (3)$$

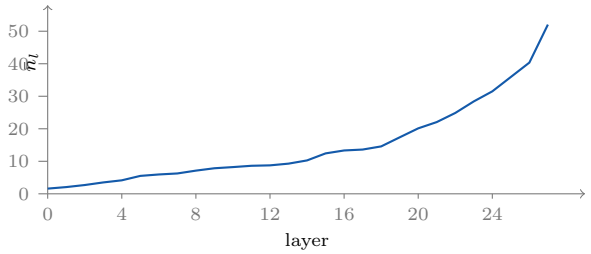


Figure 2: Mean residual norm by layer, over 256 texts. The 33-fold growth is why α is expressed as a fraction of $\bar{n}_l$ rather than in absolute units.

Pairs are generated by the model itself, prompted once per axis.

Difference PCA. The leading right singular vector of the centred per-pair differences, sign-aligned to the mean difference. This detects a mean dragged by outliers.

3.1 Scaling strength by depth

Residual norms grow with depth. Figure 2 plots the measured mean activation norm per layer: 1.60 at layer 0 rising to 52.04 at layer 27, a factor of 33. A raw α of 1.0 is therefore a small nudge early and an overwrite late, and a layer sweep on raw α compares nothing.

Each bank stores those norms and injects

$$\alpha \cdot \frac{v_l}{\|v_l\|} \cdot \bar{n}_l, \qquad \bar{n}_l = \mathbb{E}\big[\|h_l\|\big], \qquad (4)$$

so α means a fraction of a typical hidden-state magnitude and is comparable across depth. Every α in this report is that quantity.

4 A defect that faked a result

The recorder pooled a hidden state and moved it to host memory in one call: `pooled.detach().to("cpu", torch.float32)`. On the MPS backend, a combined device-and-dtype cast of a half-precision tensor *view* ignores the view's storage offset and copies from the base of the allocation.

The failure is narrow, which is why it survived review. It fires only when the view reports itself contiguous. For `hidden[:,-1,:]` that means batch size one, and batch size one was the only shape the recorder ever produced. A batched view is strided, takes a gather path, and is unaffected. Figure 3 shows the mechanism.

A minimal reproduction uses no project code. With `x = arange(60).reshape(1,12,5)` as bfloat16 on MPS, `x[:,-1,:].to('cpu', torch.float32)` yields 2.362×10^{-41} where `x[:,-1,:].float().to('cpu')` yields 55.0.

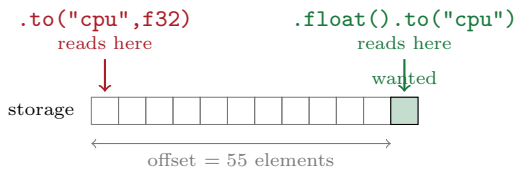


Figure 3: The view `x[:,-1,:]` begins 55 elements into its allocation. A fused device-and-dtype cast discards that offset. Casting dtype first allocates a fresh contiguous tensor at offset zero, so the host copy is correct.

Quantity	Defective	Fixed
$\|h\|$ recorded, layer 0	54.69	1.57
$\|h\|$ recorded, layer 14	55.10	12.28
$\|h\|$ recorded, layer 27	55.14	45.21
cosine vs. ground truth	0.02	1.000000
bit-identical layer pairs	11	0
axis cosine, mean absolute	0.983	0.223

Table 1: Before and after casting dtype before device. Ground truth is an independent capture through a separate hook.

The consequences were systematic, and they looked like a finding rather than a fault. Table 1 lists them. Recorded norms were identical at every layer, which no residual stream is. Eleven layer-pairs held bit-identical 3072-dimensional mean vectors over 256 texts. The structural and lexical vectors sat at cosine 0.983 at every layer, which reads as “the model encodes a single simplify direction”. That conclusion was drawn and was wrong.

Two lessons are worth separating. The arithmetic error was in a one-line utility with no branching. The detection required comparing the recorder against an independent capture of the same tensor, because every internal consistency check passed: shapes, dtypes, and finite values were all correct.

5 Other defects

Table 2 lists the remaining defects found by targeted testing. None was reported by static analysis. A linter was run over the codebase afterwards and produced 4 errors and 56 warnings, all of which were verified as false positives: its `assert-tuple` rule flags the correct `assert cond, (a, b)` two-argument form, and its bare-except rule flags clauses that specify a type.

6 Independent dials

Two unit vectors a and b with $a \cdot b = c$ are made orthogonal by Gram-Schmidt in one direction. Applying it in both directions does not orthogonalise the pair. Writing $a' = a - cb$ and $b' = b - ca$,

$$\frac{a' \cdot b'}{\|a'\| \, \|b'\|} = \frac{c - c^2 - c^2 + c^3}{1 - c^2} = -c. \tag{5}$$

The correlation changes sign and keeps its magnitude. Measured on real banks, $+0.500$ became -0.504, matching Equation 5 to three decimals.

Löwdin symmetric orthogonalisation is the correct operation. With V the matrix of axis vectors and $G = V^\top V$ its Gram matrix,

$$V' = V G^{-1/2} \tag{6}$$

gives an exactly orthonormal set, closest to V in Frobenius norm, treating every axis alike. Verified residual cosine is 2×10^{-7} on synthetic input and 1.02×10^{-7} on the real banks. A small eigenvalue of G signals that the axes were nearly collinear at that layer and the orthogonal directions are close to arbitrary, so the smallest eigenvalue is reported.

7 Metrics

Four families of measurement are needed, and the interaction between them is the part that is easy to get wrong.

Readability. Flesch-Kincaid grade level, Flesch Reading Ease, Dale-Chall hard-word rate counted per token, words per sentence, clauses per sentence, syllables per word.

Faithfulness. Bidirectional natural language inference. One direction is insufficient. Requiring only that the source entails the output rewards deleting facts, because a shorter claim is trivially entailed. Requiring only the reverse rewards inventing detail. Both are required. Scoring is per sentence so a single dropped fact appears in the minimum rather than washing out of an average.

Fluency. Perplexity of the output under the same model with no hook attached. The reference model must be untouched for the number to mean anything.

Reference-based quality. SARI, over n-grams of order 1 to 4, averaging additions, retentions and deletions. Our implementation agrees with the reference implementation to 2.84×10^{-14} over 3000 randomised cases.

7.1 Gate simplicity on fluency

Every simplicity metric is maximised by broken text. Collapsed output such as `"The,, a a the the the"`

Site	Mechanism	Consequence
Prefill detection	Prefill inferred from a call counter; decode assumed a sequence length of one	With `use_cache=False` the generated-tokens-only mode injected into every prompt position
Hook state reset	Counter reset on block entry, not per generation	Only the first generation inside one `with` block was correct; later ones either stopped steering or steered the prompt
Position validation	`positions` was an unchecked string literal	`"prompt_alll"` silently behaved as `"all"`
Gemma-3n layer output	`altup` returns `[streams, batch, seq, hidden]`, rank 4	Three-dimensional indexing addressed batch instead of sequence on the whole Gemma path
Prompt mean pooling	Attention mask never reaches a forward hook	Averaged over left padding
Local model resolution	Directory judged complete once `config.json` existed	Truncated shards were loaded; a later fix matched its own `.aria2` *directory* and refused every load
Pad token	`pad_token_id or eos_token_id`	Gemma-3n has pad id 0, so the falsy test replaced it with the end-of-sequence token
Perplexity masking	Prompt and prompt-plus-output tokenised separately	BPE merges across the join shifted every label; masking now uses the shared prefix
Hard-word rate	`textstat.difficult_words` counts *distinct* hard words	Reusing one hard term read as a 50% lexical simplification. Two 15-word texts scored 2 against 4 deduplicated and 4 against 4 per token

Table 2: Defects found by differential testing. Each altered a measured quantity.

contains no difficult words and no long sentences. One sweep cell recorded a 91% reduction in hard-word rate with every output destroyed. Simplicity is therefore credited only on outputs that pass a degeneracy check, and the number of survivors is reported alongside.

7.2 Why the check is narrow

Genuine simplification is statistically similar to degeneration. The valid sentence group *The cell makes energy. The cell uses energy. The cell needs energy.* has a type-token ratio of 0.5 and 45% of its bigrams repeated. Those are the same signals a broken generation produces, and that text is exactly what the system is meant to output.

Two plausible backstops do not work, and both were tested rather than assumed.

Perplexity does not detect repetition. Repetition is low-entropy, so a loop is cheaper under the model than ordinary prose. The broken fragment *organe-cepted control-cepted control-cepted contr* scored 18.8 while the valid repetitive group above scored 37.0. Any threshold catching the first rejects the second. Perplexity guards high-entropy breakage only.

Dictionary lookup does not work either. The system word list is Webster's Second, which predates most scientific vocabulary, so *mitochondrion* reads as out-of-vocabulary. Valid academic prose scored 0.22 out-of-vocabulary against 0.27 for the broken fragment.

What remains is a rule set that fires only on text that is not language. Figure 4 shows the order of evaluation.

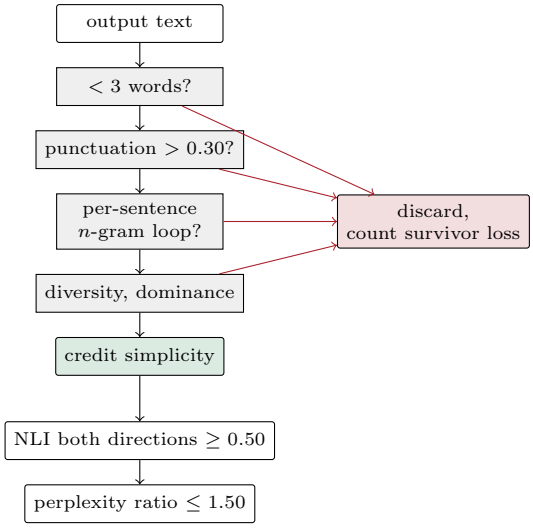


Figure 4: Gate order. Degeneracy is checked before any simplicity metric is credited, because broken text maximises all of them.

Immediate n-gram repetition is the discriminating signal, with two corrections. It must be computed per sentence, because a flat token list cannot see sentence boundaries and the ordinary pair *. . . sleet and freezing rain. Freezing rain can be heavy. . .* then reads as a repeating bigram. It must also exclude loops built from function words, because *on and on and on* and *row after row after row* are English while *cepted control cepted control* is not, unless the phrase repeats three or more times. The final rule set produces no false positives and no false negatives across 13 hand-built cases and 24 real paragraphs.

7.3 FKGL fails on sentences

Flesch-Kincaid is a passage-level formula. At single-sentence length it is dominated by syllables per word and by the sentence's own length, so it rewards padding and penalises concision. This is not a small effect. A faithful 17-word paraphrase with NLI 0.990 against its source scored 14.3, while a 60-word rewrite that invented three entities and scored NLI 0.090 scored 7.0.

The consequence for this project is direct: an early comparison used Flesch-Kincaid to conclude that prompting outperformed steering. That comparison measured padding. Sentence-level judgements belong to SARI and to the faithfulness gate.

8 Data

ASSET supplies 359 test and 2000 validation sentences with 10 human references each. The 10 references are what makes SARI meaningful, so all are retained.

The estimation corpus is 256 texts drawn from a 665-item filtered pool of ASSET validation sources. Disjointness from the test set is recomputed on every call rather than trusted from the upstream split, so a stale cache cannot reintroduce leakage. Passages come from a pool of 11 854 paragraphs, of which 2 were dropped for containing an ASSET sentence verbatim.

OneStopEnglish required care. The flat three-class datasets are not index-aligned, although they appear to be. Content-overlap matching identified the presumed partner for only 184, 186 and 187 of 189 pairs across the three level combinations, and index 53 pairs a skydiving article with a royal birth. A duplicated elementary text shifts everything after it. Aligned triples were instead taken from OneStopQA, which carries explicit title, paragraph and level keys, yielding 162 aligned paragraph triples.

9 Layer and strength sweep

The full grid of 28 layers, 8 strengths, 2 axes and 48 texts is roughly a day of wall clock. It is split. Stage 1 scans every layer at two coarse strengths with 12 texts. Stage 2 runs the full strength grid on surviving layers and adds NLI.

Effects are measured against each text's own *unsteered rewrite*, not against the source. Both are rewrites, so the difference isolates what steering did from what asking for a rewrite did.

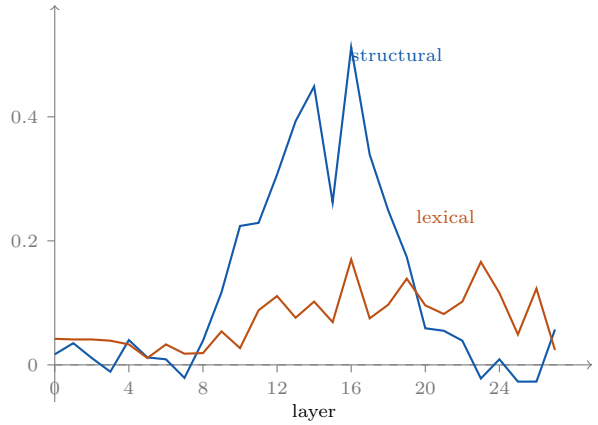


Figure 5: Stage-1 own-axis effect by layer, as a fractional reduction against the unsteered rewrite, taking the better of the two coarse strengths. Structural effect is confined to layers 10 to 18. Values at layers 14 and 16 rest on 9 and 8 surviving outputs of 12 and are not used for selection.

9.1 Depth

Figure 5 plots stage-1 results. The structural response is confined to layers 10 to 18 and is absent on either side. The lexical response is smaller and spread, with local maxima at layers 16 and 23.

Axis entanglement depends strongly on the estimator. Figure 6 plots per-layer cosine for both. Instruction difference gives mean absolute cosine 0.223; contrastive activation addition gives 0.673, reaching 0.976 at layer 1.

The reason is structural, not a data-quality problem. Inspection confirms the generated pairs are well isolated: for a source about Gothic architecture, the lexical rewrite preserved sentence structure and substituted *churches*, *monasteries* and *village churches*, while the structural rewrite split the sentence into three. Contrastive addition nevertheless conflates the axes because it contrasts an output against its source, and both axis outputs share the property being varied, namely being simpler than the source. Differencing two instructions that differ along one axis cancels that shared component instead.

9.2 Strength

Figure 7 plots stage-2 response. Three properties matter.

Response is not monotone. Structural effect at layer 17 peaks at $+0.339$ at $\alpha = 1.0$ and falls to $+0.027$ at $\alpha = 1.25$ as degeneration begins. Any search over α must therefore bisect inside a bracket verified to be both monotone and gate-passing, never assume monotonicity globally.

The sign is correct. At $\alpha = -1.0$ the structural vector

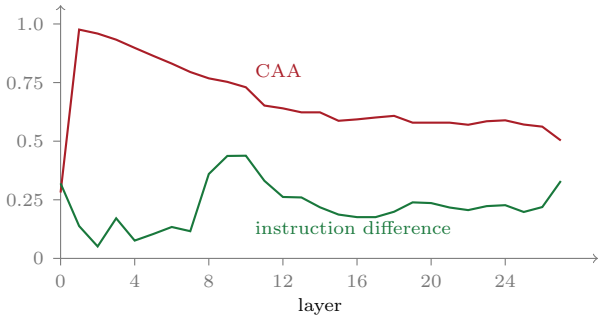


Figure 6: Absolute cosine between the structural and lexical vectors, per layer, by estimator. Lower is more separable.

Axis	Site	Own	Other	NLI	Gate
structural	L17, 1.00	+0.339	+0.070	0.701	75%
structural	L15, 0.75	+0.218	−0.014	0.765	83%
structural	L15, 0.50	+0.204	−0.010	**0.873**	**100%**
lexical	L19, 1.00	+0.139	−0.105	0.830	92%
lexical	L26, 1.00	+0.123	−0.178	0.749	83%
lexical	L19, 0.75	+0.111	−0.021	0.853	92%

Table 3: Best cells by own-axis effect. Structural is both stronger and more selective: at L15 with $\alpha = 0.5$ it moves its own axis by +0.204 while disturbing the other by −0.010. Lexical always lengthens sentences.

at layer 17 *lengthens* sentences by 60%, own-axis effect −0.598. A direction that simplified in both directions would indicate an artifact rather than a concept.

The two axes have different usable ranges. Structural saturates and then collapses between 1.0 and 1.25, while lexical is still climbing at 1.5. Stage 1 sampled only 0.5 and 1.0 and consequently reported the lexical axis as weak; that was a property of the grid, not of the axis.

9.3 A metric that rewarded inaction

Selectivity was defined as own-axis effect minus the absolute cross-axis effect. A cell with no effect at all scores $0 - 0 = 0$, which outranks a cell with a large own-axis effect and a moderate cross-axis one. The first stage-1 report consequently named layer 4 the best lexical site at selectivity +0.002, where both effects were noise. Selection now requires a minimum own-axis effect of 0.05 before selectivity is considered.

Table 3 lists the best cells under the corrected rule, at zero degeneration and with the faithfulness gate passing on at least 75% of items.

10 Calibration

Mapping α onto a named reading level requires searching for the α that reaches a target Flesch-Kincaid grade subject to the gates. No target was reachable. Gate-passing grade spans 12.19 to 14.76 for the structural axis and 13.84 to 14.76 for the lexical axis, against a baseline of 14.76 and targets of 4, 6 and 8.

Pushing harder does not help, because there is no range left to spend. At $\alpha = 1.389$ the grade *rises* to 15.77 with NLI 0.385 and 25% of outputs degenerate. At $\alpha = 2.0$ the grade is 21.61, NLI 0.153, and 75% are degenerate. The search reports this as unreachable rather than returning the closest gate-violating point.

Configuration	FKGL	w/sent	NLI	Degen
unsteered	16.80	27.5	–	0%
prompt-only	**7.91**	17.1	**0.161**	0%
L15, α=0.75	13.74	21.6	0.691	0%
L12–18, α=0.1	13.64	20.2	**0.847**	0%
L10–19, α=0.1	**12.79**	19.6	0.595	0%
L12–18, α=0.2	10.02	11.6	0.834	**83%**

Table 4: Band probe, 12 held-out texts. Prompt-only reaches the lowest grade and fails the faithfulness gate by a wide margin.

11 Layer bands

Spreading a smaller α across a band of layers is the obvious way to widen the range. It buys little and costs stability. Table 4 gives the measurements. The best zero-degeneration band reaches grade 12.79 against 13.74 for the best single layer, an improvement of about one grade level. The cliff becomes far steeper: the same band moves from 0% to 83% degenerate between $\alpha = 0.1$ and 0.2. Bands across both axes collapse entirely. Single-layer injection with a calibrated α remains the better operating point.

12 Evaluation

Table 5 reports all six variants on the ASSET test set at $n = 32$, with structural steering at layer 15 and $\alpha = 0.75$ and lexical steering at layer 19 and $\alpha = 1.0$.

Aggregate means can hide item-level variance, so every comparison is paired per item. Figure 8 plots mean SARI difference against the unsteered baseline with 95% confidence intervals.

Three results follow.

Structural steering beats prompt-only significantly. On the same paired basis, SARI improves by +5.47 (CI +2.45 to +8.49, 23 wins against 9) and faithfulness by +0.36 (CI +0.23 to +0.50, 27 against 5).

Structural steering does not beat doing nothing on SARI. The difference is +0.43 with an interval spanning zero. What it does is reduce sentence length

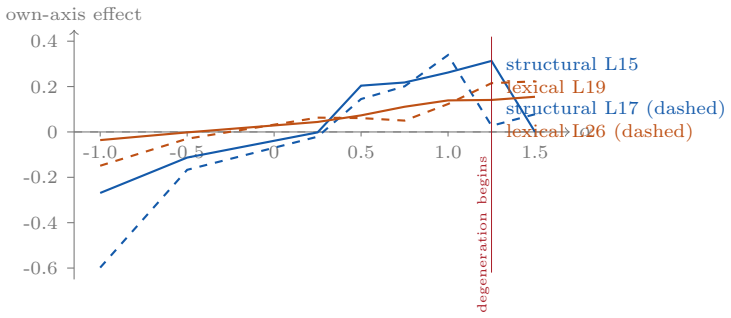


Figure 7: Stage-2 strength response. Negative α reverses the effect, which confirms the vectors encode the intended concepts. The structural curves turn over past $\alpha = 1.0$; the lexical curves are still rising at 1.5.

Variant	FKGL	w/sent	hard	NLI	Gate	PPL ratio	SARI	Degen
unsteered	12.49	20.36	0.345	0.875	90.6%	1.00	45.44	0%
prompted	**6.78**	15.15	0.161	0.431	40.6%	3.64	40.40	0%
steered structural	10.15	16.62	0.323	0.795	81.2%	1.31	**45.87**	**0%**
steered lexical	12.30	21.84	0.324	0.845	90.6%	1.20	44.07	0%
steered dual	13.72	30.85	0.234	0.579	50.0%	2.02	41.17	34.4%
prompted + steered	5.32	15.39	0.095	0.225	15.6%	4.74	31.94	65.6%

Table 5: ASSET test, $n = 32$, 10 references per item. Prompting reaches the lowest grade level and the worst faithfulness. Perplexity ratio above 1.50 fails the fluency gate.

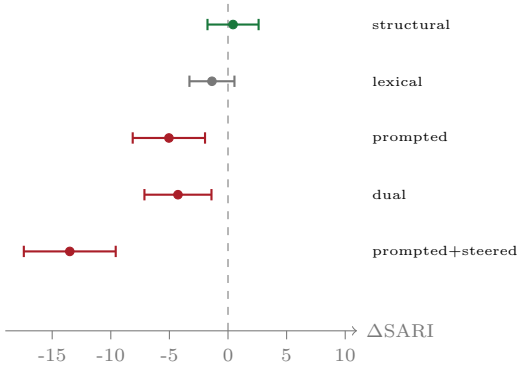


Figure 8: Paired per-item SARI difference against no intervention, 95% intervals, $n = 32$. Structural steering is a tie: its interval spans zero. The three lower rows are significantly worse than no intervention. Only structural steering fails to be worse.

18% and grade level 19% at no measurable cost in SARI and none in faithfulness (−0.08, CI −0.21 to +0.05). That satisfies the requirement, which was to simplify without changing meaning, not to improve on the source.

Prompting is significantly worse than no intervention, at −5.04 SARI with NLI 0.431 and a perplexity ratio of 3.64.

12.1 Output samples

Table 6 shows why the faithfulness numbers diverge from the grade levels. Prompting reaches a low grade by replacing content. In the second case it substitutes an analogy for a statistic and drops the figure entirely.

13 Conclusions

The two axes are separable directions at mean absolute cosine 0.223 and they respond at different depths, structure at layers 15 to 17 and vocabulary at layers 19 to 26. The two-dial hypothesis holds as a statement about mechanism.

It does not hold as a product. The lexical dial is a SARI tie trending negative at −1.37, it lengthens sentences as a side effect, and running both dials together degenerates 34% of outputs. The deliverable is one structural dial.

Steering is a modulator, not a replacement for a prompt. It cannot reach a target reading level alone. Composition with a prompt is destructive rather than additive: that variant is the worst tested, at −13.50 SARI and 66% degenerate.

What steering does deliver is a continuous strength parameter that a prompt cannot provide, and better meaning preservation than prompting at any setting tested.

14 Limitations

Results are $n = 32$ on one evaluation set. The contrastive and PCA estimators were compared for axis separation only and never evaluated as steering vectors. Passage-level results are $n \leq 4$. Sentence-by-sentence

	Case 1	Case 2
Source	Some academic journals do refer to Wikipedia articles, but are not elevating it to the same level as traditional references.	The aggregate income distribution is highly concentrated towards the top, with the top 6.37% earning roughly one third of all income.
Prompt-only	You might have noticed that some big schools and universities use Wikipedia articles in their research. . . they don't think it's as good as other sources, like books or articles from experts. FKGL 7.0, NLI **0.090**	Imagine you have a big jar of cookies, and you want to know how the cookies are divided among the people. . . some groups have a lot more cookies than others. NLI **0.025**
Steered, L15, α=0.75	Some academic journals mention Wikipedia articles, but they do not consider them equal to traditional references. FKGL 14.3, NLI **0.990**	The US income distribution is highly skewed towards the top, with the top 6.37% of earners holding about one-third of the country's total income. no-hallucination **0.63**

Table 6: Prompting invents universities, books and experts in case 1, and replaces a statistic with a cookie-jar analogy in case 2, dropping the 6.37% figure. Steering retains both. Case 1 also shows the Flesch-Kincaid inversion: the faithful output scores worse.

passage simplification is unreliable at this model size: on one paragraph it emitted the same sentence 6 times of 9 and invented a date not present in the source, so context is disabled by default. Porting to Gemma-3n requires a fresh sweep, because its decoder returns rank-4 output and the layer indices do not transfer.

Appendix: concepts used above

Residual stream

A decoder-only transformer keeps one vector per token position, of width equal to the hidden size, here 3072. Each layer reads that vector, computes attention and a feed-forward update, and *adds* the result back. Nothing is overwritten. The stream is therefore a running sum, and a direction added at layer l persists through every later layer unless something cancels it. This is what makes steering possible with a single addition.

Forward hook

A callback attached to a module that fires after its forward pass, receiving the module, its inputs and its output. Returning `None` leaves the output alone; returning a tensor replaces it. Reading uses the first form, steering the second. A hook that accidentally returns a value replaces the layer output with it, which is a common source of confusing failures.

Activation steering and CAA

Activation steering adds a fixed direction to internal activations at inference to change behaviour, with no weight updates. Contrastive Activation Addition obtains that direction as the mean difference in activations between paired inputs that differ in the target property: $v = \mathbb{E}[h^+] - \mathbb{E}[h^-]$. The assumption is that the property is represented approximately linearly, so a single direction suffices.

bfloat16 and MPS

bfloat16 keeps the 8 exponent bits of float32 and 8 mantissa bits, so it has the dynamic range of float32 and about 3 significant decimal digits. MPS is PyTorch's Metal backend for Apple GPUs. Reduced-precision paths on newer backends are less exercised than CUDA equivalents, which is the class of defect described in Section 4.

Views, strides and storage offset

A tensor is a view onto a flat allocation, described by a shape, a stride per dimension, and a storage offset. `x[:,-1,:]` does not copy: it points into the same allocation at offset $(\text{seq}-1)\times\text{hidden}$. A view is called contiguous when its strides match a dense row-major layout. Contiguity says nothing about the offset, and a copy routine that assumes offset zero for contiguous inputs reads the wrong memory. That is the exact failure in Section 4.

Flesch-Kincaid grade level

$$0.39\frac{W}{S} + 11.8\frac{Y}{W} - 15.59$$

for W words, S sentences and Y syllables. It reports a US school grade. Both terms grow with length, so on a single sentence the first term is the sentence's own word count and the formula rewards splitting and padding. Flesch Reading Ease is the same inputs on a 0 to 100 scale, higher being easier.

Dale-Chall

Uses a list of about 3000 words familiar to fourth-grade readers. Anything off the list is a hard word. The rate must be counted per token: counting distinct hard words means a text reusing one hard term scores easier than one using several, which inverts the measurement for technical prose that repeats its key terms.

SARI

The standard simplification metric, comparing system output against the source *and* multiple human references over n-grams of order 1 to 4. It scores three operations. Additions are n-grams in the output but not the source, credited when references also add them. Retentions are n-grams kept from the source, credited when references keep them. Deletions are n-grams removed from the source, credited when references remove them. Additions and retentions use F1; deletions use precision only, because penalising a system for retaining material that some reference deleted proved unreliable. The final score averages the three, each averaged over the four n-gram orders. Counts are weighted by how many references contain each n-gram, so a set-membership implementation gives wrong numbers.

Natural language inference

Given a premise and a hypothesis, classify their relation as entailment, neutral or contradiction. MNLI is the standard multi-genre training corpus, and `roberta-large-mnli` is a model fine-tuned on it. Entailment probability is used here as a faithfulness proxy.

Direction matters. If the source entails the output, the output adds nothing false, but it may have dropped half the content: a shorter claim is easier to entail. If the output entails the source, nothing was dropped, but material may have been invented. Requiring both bounds omission and hallucination simultaneously. The known weakness is that NLI models penalise heavy paraphrase, so a low score is evidence of either information loss or aggressive rewording, and the two are not separated by the score alone.

Perplexity

exp of the mean negative log-likelihood per token. Conditional perplexity scores only the continuation given a prompt. Lower means the model finds the text more predictable. Predictable is not the same as good: repetition is highly predictable, so looping text has *low* perplexity. Perplexity therefore detects high-entropy breakage such as

random tokens, and misses low-entropy breakage such as loops.

Distinct-n and type-token ratio

Distinct-n is the count of unique n-grams over the total, a diversity measure. Type-token ratio is distinct-1 over word count. Both fall when text degenerates, and both also fall when text is legitimately simplified, which is why neither can serve as a degeneracy test on its own.

Gram-Schmidt and Löwdin

Gram-Schmidt removes from one vector its projection onto another: $a' = a - (a \cdot b)b$. It is asymmetric, leaving b untouched. Applying it in both directions does not produce an orthogonal pair, as Equation 5 shows. Löwdin symmetric orthogonalisation multiplies the whole set by $G^{-1/2}$ for Gram matrix G, producing an exactly orthonormal set that is the closest such set to the original in Frobenius norm, with no privileged member. $G^{-1/2}$ is computed from the eigendecomposition of G, which is symmetric positive semi-definite.

PCA in this context

Given per-pair difference vectors, principal component analysis finds the direction of greatest variance after centring. If the leading component and the plain mean disagree, the mean is being pulled by a few outlying pairs. Sign is ambiguous in PCA, so the component is aligned to the mean before use.

Gemma-3n, altup and MatFormer

Gemma-3n uses alternating updates, which maintain several parallel prediction streams through the stack rather than one residual stream, so a decoder layer returns a rank-4 tensor with streams on the leading axis. MatFormer denotes nested submodels of different widths inside one set of weights. Both mean that code written against a single rank-3 residual stream is wrong there, and that layer indices found on one architecture carry no meaning on the other.

LoRA, for contrast

Low-rank adaptation trains small matrices A and B so that an update $\Delta W = BA$ is added to frozen weights. It changes the model permanently and requires training. Steering changes nothing and requires none, and its strength is adjustable at inference. The trade is capability: steering can only move along directions the model already represents.

safetensors

A tensor container format storing a JSON header followed by raw tensor bytes. It supports memory-mapped reads and carries no executable code, unlike pickle. The header holds a string-to-string metadata map, used here to record which estimator produced a bank and whether it was orthogonalised.

aria2c

A download utility that opens several concurrent connections per file and can fetch several files at once. A single HTTPS stream is usually latency-bound well below the available bandwidth, so 16 connections per file is materially faster than one. The authorisation header is passed through the input file rather than the command line so it does not appear in the process list.

Paired testing and confidence intervals

Comparing two systems by their mean scores discards the pairing, and per-item variance in simplification is large. Pairing instead computes a difference per item and tests whether the mean difference differs from zero. A 95% confidence interval that spans zero means the data do not distinguish the two systems. This is why +0.43 SARI with interval −1.75 to +2.61 is reported as a tie and not as an improvement.